Department of Information Technology
Academic Year 2025 - 2026 (Odd Semester)

Semester, Degree & Branch: V Semester, B.Tech. IT
Course Code & Title: CS3551 & Distributed Computing
Name of the Faculty member (s): Mrs. A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Snapshot Algorithms for FIFO Channels
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen: Crossword Puzzle**
- **Justification:**

  The concept of Snapshot Algorithms for FIFO Channels (Chandy-Lamport Algorithm) has been explained thoroughly, including the importance of consistent global state, marker sending/receiving rules, and applications in distributed systems like checkpointing, recovery, and deadlock detection. To evaluate the understanding of students, a crossword puzzle was chosen as an innovative method. Crossword puzzles encourage critical thinking and improve technical vocabulary, problem-solving, and memory recall. They also provide an engaging and stress-free way for students to revise distributed computing concepts while promoting teamwork and active learning.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
  - At the completion of Unit II**,** all topics related to Snapshot Algorithms were revised.
  - The time allotted for the activity was 10 minutes**.**
  - Students were divided into two groups: boys group and girls group**.**
  - Crossword Puzzle 1 (based on definitions, rules, and properties of Snapshot Algorithms) was displayed for 5 minutes, and a representative from the boys' group filled the answers on the board with the support of their team.
  - Crossword Puzzle 2 was displayed for the girls group, and similarly, one representative filled the answers based on team discussion.
  - Both groups completed the activity successfully.
  - Sample activity screenshots are shown in Figure 1 and Figure 2**.**

- **CO – PO / PSO mapping:**

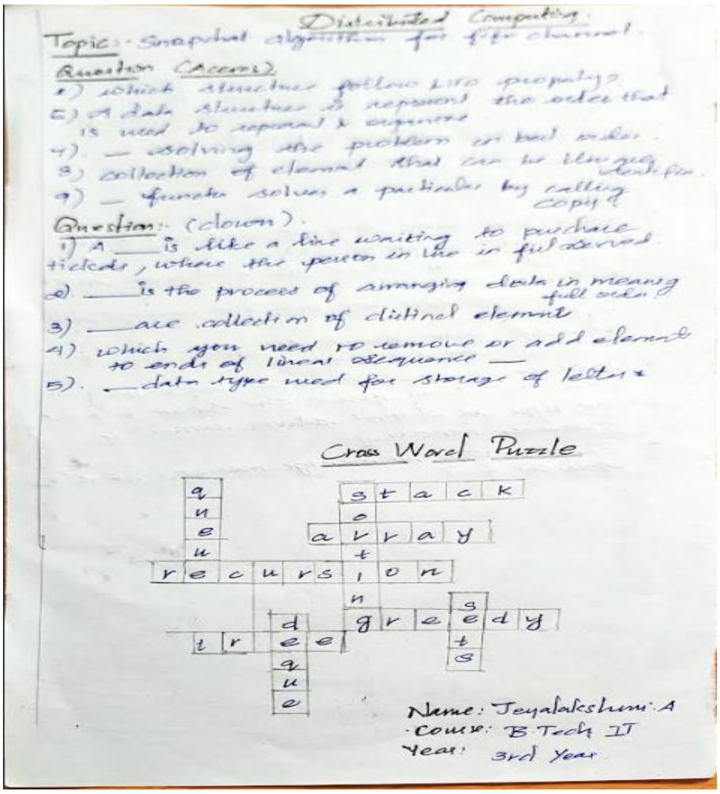
CO	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
CO1	**3**	**2**	**1**	**1**	**2**	**2**	**2**	**3**	**3**	**2**

(1 – Low 2 – Moderate 3 – High)

- **PO/PSO Mapped:**

Innovatie Practice	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
	3	**2**	**1**	**1**	**2**	**2**	**2**	**3**	**3**	**2**
Justification for Correlation	Gain knowledge of group communication, global state, and snapshot recording algorithms using basic IT concepts.	Identify and analyze global state and snapshot recording concepts to formulate suitable solutions for IT-related problems.	Apply the knowledge of snapshot algorithms and global state tracking to provide solutions for complex distributed system problems.	Use the knowledge of group communication, global state mechanisms, and snapshot recording to analyze scenarios in distributed computing.	Perform causal ordering and group communication while adhering to ethical principles and rules in distributed systems.	Implement message ordering and snapshot recording mechanisms in distributed computing applications.	Continuously update knowledge in distributed computing to adopt emerging snapshot algorithms and consistency models.	Develop distributed software by implementing snapshot algorithms, global state tracking, and causal ordering mechanisms.	Provide efficient distributed solutions using knowledge of message ordering and snapshot recording.	Identify, evaluate, and apply suitable distributed computing approaches for group communication and state recording.

- **Images / Screenshot of the practice:**



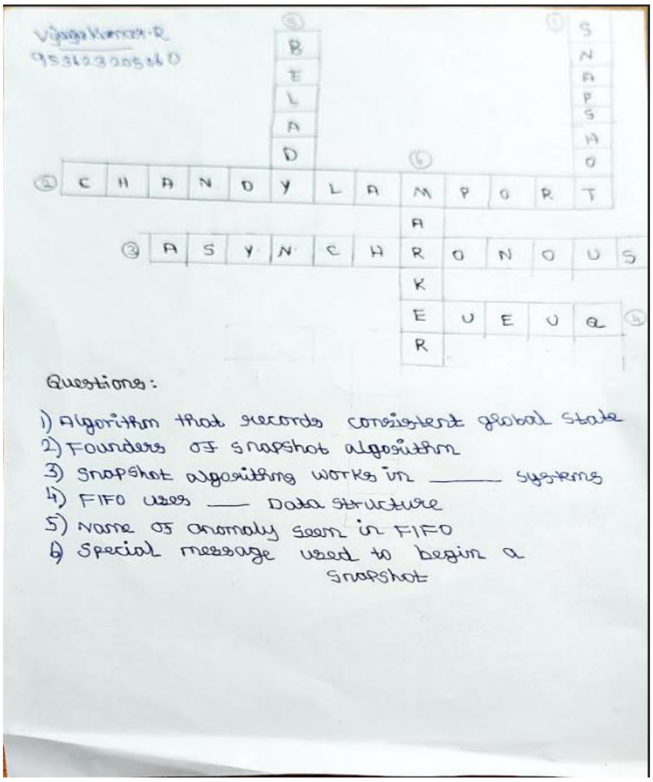


Figure 1. a Puzzle solved by Girls Group

Figure 1. b Puzzle solved by Boys Group

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - Some of the students felt it was easy to recall and identify answers for the given crossword questions related to snapshot algorithms.
  - A few students answered enthusiastically, while some did not participate much even though they were aware of the concepts.
- ***Benefit of the practice:***
  - Students actively participated in the puzzle activity and revised the concepts of global state, markers, and channel state in a fun way.
  - The activity helped students discuss and collaborate with their peers to arrive at the correct answers.
  - The method encouraged teamwork and improved understanding of distributed computing concepts through interactive learning.
- ***Challenges faced in implementation:***
  - A few students did not show much interest in solving the crossword.

- Some students struggled with recalling concepts like marker receiving rule and channel state recording due to a lack of conceptual clarity.
- After the final discussion and explanation, they gained a better understanding of snapshot algorithms and learned new technical terms.

References:

- https://crosswordlabs.com/
- https://wordmint.com/
- https://puzzlemaker.discoveryeducation.com/criss-cross
- https://worksheets.theteacherscorner.net/make-your-own/crossword

Signature of Faculty Member **HOD**